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Semiconductor package

Abstract

In a semiconductor package, flow guiding strips are provided on a guiding area of a flexible substrate to separate a chip and the flexible substrate such that a filling material flowing between the chip and the flexible substrate can squeeze out the air between the chip and the flexible substrate to improve the reliability of the semiconductor package.

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Field of Classification Search

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10325783	12/2018	Munding	N/A	H01L 25/0657
2010/0038780	12/2009	Daubenspeck	257/E23.129	H10D 62/117
2019/0103375	12/2018	Huang	N/A	H01L 24/19

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
9-232474	12/1996	JP	N/A
2006-13030	12/2005	JP	N/A
2007-173738	12/2006	JP	N/A
WO-2009050891	12/2008	WO	H01L 21/563

OTHER PUBLICATIONS

Korean Office Action mailed Feb. 22, 2024 for Korean utility model Application No. 20-2022-0002544, 7 pages. cited by applicant

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Background/Summary

FIELD OF THE INVENTION

(1) This invention relates to a semiconductor package, and more particularly to a semiconductor package with an improved reliability by helping a filling material to squeeze out air between chip and flexible substrate.

BACKGROUND OF THE INVENTION

(2) In conventional technology, conductive pads are arranged around a chip's surface and the chip is electrically connected to a substrate by bumps disposed on the conductive pads. In order to enhance the bonding strength of the chip to the substrate and prevent oxidation of the bumps, an underfill is dispensed along the chip's edge to flow by capillarity and fill the space between the chip and the substrate. However, an area without bumps between the chip and the substrate may slow down the

flow of the underfill, fail to squeeze out the air between the chip and the substrate and lead the air to be covered by the underfill between the chip and the substrate. Furthermore, the area without the bumps may cause uncontrollable deformation of the substrate, such as warpage. While the deformed substrate contacts the chip, the underfill may be blocked to cover the air between the chip and the substrate and reduce filling rate and package reliability.

SUMMARY

(3) One object of the present invention is to provide a semiconductor package which has flow guiding strips disposed on a guiding area of a flexible substrate. Because of the flow guiding strips, a filling material flowing between a chip and the flexible substrate can squeeze out the air between the chip and the flexible substrate to improve reliability of the semiconductor package.

(4) A semiconductor package of the present invention includes a chip, bumps and a flexible substrate. The chip includes conductive pads which are visible from the chip's first surface. A first area and a second area are defined on the first surface, the first area is located between an edge of the first surface and the second area, the conductive pads are arranged on the first area. The bumps are disposed on the conductive pads respectively. The flexible substrate includes conductive lines and flow guiding strips. A chip mounting area, a guiding area and a wire area are defined on a second surface of the flexible substrate, the guiding area is located inside the chip mounting area. Each of the conductive lines includes an inner lead and a wire which are connected with each other. Along a first direction, the inner lead is located on the chip mounting area, the wire is located on the wire area, and the flow guiding strips are located on the guiding area. The flow guiding strips are extended toward the inner lead of each of the conductive lines along the first direction and not connected to the conductive lines. A first guiding channel is located between the adjacent flow guiding strips along the first direction, a second guiding channel is located between the adjacent flow guiding strips along a second direction which intersects the first direction, and the second guiding channel is communicated with the first guiding channel. The chip is mounted on the chip mounting area and bonded to the inner lead of each of the conductive lines by the bumps, the first surface of the chip is faced toward the second surface of the flexible substrate. The second area is located above the guiding area, the flow guiding strips are located between the second area and the guiding area and provided to separate the first surface of the chip and the second surface of the flexible substrate.

(5) Owing to the flow guiding strips disposed on the guiding area and the first and second guiding channels located between the adjacent flow guiding strips, the flowing filling material can squeeze out the air between the chip and the flexible substrate to increase filling rate of the filling material.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-section view diagram illustrating a semiconductor package in accordance with one embodiment of the present invention.

(2) FIG. 2 is a top view diagram illustrating a chip of a semiconductor package in accordance with one embodiment of the present invention.

(3) FIG. 3 is a top view diagram illustrating a part of a flexible substrate of a semiconductor package in accordance with one embodiment of the present invention.

(4) FIG. 4 is a top view diagram illustrating a part of a flexible substrate of a semiconductor package in accordance with another embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(5) With reference to FIGS. 1 to 3, a semiconductor package **100** in accordance with one embodiment of the present invention includes a chip **110**, a plurality of bumps **120**, a flexible substrate **130** and a filling material **140**. The chip **110** includes a first surface **111** and a plurality of

conductive pads **112** which are visible from the first surface **111**. A first area **111b** and a second area **111c** are defined on the first surface **111**, the first area **111b** is located between an edge **111a** of the first surface **111** and the second area **111c**. The conductive pads **112** are located on the first area **111b**, and the bumps **120** are disposed on the conductive pads **112**, respectively.

(6) With reference to FIGS. **1** and **3**, the flexible substrate **130** includes a second surface **131**, a plurality of conductive lines **132** and a plurality of flow guiding strips **133**. A chip mounting area **131a**, a guiding area **131b** and a wire area **131c** are defined on the second surface **131**, the guiding area **131b** is located inside the chip mounting area **131a**. Each of the conductive lines **132** includes an inner lead **132a**, a wire **132b** and an outer lead (not shown), the inner lead **132a** and the outer lead are respectively located on both ends of the wire **132b** and connected to the wire **132b**. Preferably, the flexible substrate **130** further includes a solder resist layer **150** used to cover the second surface **131** of the flexible substrate **130** and the wire **132b** of each of conductive lines **132**. The inner lead **132a** and the outer lead of each of the conductive lines **132**, the guiding area **131b** and the chip mounting area **131a** on the second surface **131** of the flexible substrate **130** are not covered by the solder resist layer **150**.

(7) With reference to FIG. **3**, the guiding area **131b** has a length L greater than or equal to $300\text{ }\mu\text{m}$ along a first direction Y and has a width W greater than or equal to $300\text{ }\mu\text{m}$ along a second direction X which intersects the first direction Y . Preferably the width W of the guiding area **131b** is greater than $500\text{ }\mu\text{m}$.

(8) With reference to FIGS. **1** and **3**, the inner leads **132a** of the conductive lines **132** are located on the chip mounting area **131a** and close to the guiding area **131b**, the wires **132b** of the conductive lines **132** are located on the wire area **131c**, the flow guiding strips **133** are located on the guiding area **131b** and extended toward the inner leads **132a** of the conductive lines **132** along the first direction Y , but the flow guiding strips **133** are not connected to the conductive lines **132**.

Preferably, there is a distance D between each of the flow guiding strips **133** and the inner lead **132a** of each of the conductive lines **132**, the distance D is greater than or equal to $50\text{ }\mu\text{m}$ and is less than or equal to $100\text{ }\mu\text{m}$ along the first direction Y ($50\text{ }\mu\text{m} \leq D \leq 100\text{ }\mu\text{m}$). In this embodiment, the flow guiding strips **133** and the conductive lines **132** are made of the same material owing to they are formed together after patterning a metal layer. In other embodiments, the flow guiding strips **133** and the solder resist layer **150** are made of the same material and formed by screen printing.

(9) With reference to FIG. **3**, each of the flow guiding strips **133** has a length $L3$ greater than or equal to $100\text{ }\mu\text{m}$ and less than or equal to $300\text{ }\mu\text{m}$ along the first direction Y ($100\text{ }\mu\text{m} \leq L3 \leq 300\text{ }\mu\text{m}$), and each of the flow guiding strips **133** has a width $W3$ greater than or equal to $7\text{ }\mu\text{m}$ and less than or equal to $30\text{ }\mu\text{m}$ along the second direction X ($7\text{ }\mu\text{m} \leq W3 \leq 30\text{ }\mu\text{m}$). Along the first direction Y , a first guiding channel **R1** exists between the two adjacent flow guiding strips **133** and it has a first width $W1$ greater than or equal to $50\text{ }\mu\text{m}$ and less than or equal to $100\text{ }\mu\text{m}$ ($50\text{ }\mu\text{m} \leq W1 \leq 100\text{ }\mu\text{m}$). And along the second direction X , a second guiding channel **R2** exists between the two adjacent flow guiding strips **133** and it has a second width $W2$ greater than or equal to $100\text{ }\mu\text{m}$ and less than or equal to $500\text{ }\mu\text{m}$ ($100\text{ }\mu\text{m} \leq W2 \leq 500\text{ }\mu\text{m}$). The second guiding channel **R2** is communicated with the first guiding channel **R1**.

(10) With reference to FIGS. **1** to **3**, the chip **110** is mounted on the chip mounting area **131a** and its first surface **111** faces toward the second surface **131** of the flexible substrate **130**. The chip **110** is bonded to the inner leads **132a** of the conductive lines **132** by the bumps **120**. After bonding, the second area **111c** of the first surface **111** of the chip **110** is located above the guiding area **131b** of the second surface **131** of the flexible substrate **130**, and the flow guiding strips **133** are located between the second area **111c** and the guiding area **131b**. The flow guiding strips **133** are provided to separate the first surface **111** of the chip **110** and the second surface **131** of the flexible substrate **130**. In this embodiment, a gap G exists between the first surface **111** of the chip **110** and each of the flow guiding strips **133**. While dispensing the filling material **140** along the edge of the chip

110, the filling material **140** can flow by capillarity to fill the space between the first surface **111** of the chip **110** and the second surface **131** of the flexible substrate **130**. The filling material **140** is guided by the flow guiding strips **133**, the first guiding channel **R1** and the second guiding channel **R2** to flow through the guiding area **131b** and squeeze the air between the chip **110** and the flexible substrate **130** out of the chip mounting area **131a**. As a result, the filling material **140** can flow in the first guiding channel **R1**, the second guiding channel **R2** and the gap **G** smoothly to increase filling rate.

(11) With reference to FIG. 1, while the flexible substrate **130** has a deformation due to material property or temperature variation, the flow guiding strips **133** on the flexible substrate **130** may contact the first surface **111** of the chip **110** so as to prevent the second surface **131** of the flexible substrate **130** from contacting the first surface **111** of the chip **110**, thus the flow of the filling material **140** between the chip **110** and the flexible substrate **130** will not be blocked. The filling material **140** can smoothly fill the space between the first surface **111** of the chip **110** and the second surface **131** of the flexible substrate **130**, squeeze out the air between the chip **110** and the flexible substrate **130** and fill in the first guiding channel **R1** and the second guiding channel **R2**, consequently, the reliability of the semiconductor package **100** can be improved.

(12) FIG. 4 shows another embodiment of the present invention. In this embodiment, the flexible substrate **130** further includes at least one connective line **134** which is located on the chip mounting area **131a**, both ends **134a** of the conductive line **134** are used to bridge two of the conductive lines **132**. The connective line **134** and the inner leads **132a** of the conductive lines **132** surround the guiding area **131b**. While dispensing the filling material **140** along the edge of the chip **110**, the filling material **140** can flow by capillarity to fill the space between the first surface **111** of the chip **110** and the second surface **131** of the flexible substrate **130**. By the guiding of the flow guiding strips **133**, the first guiding channel **R1** and the second guiding channel **R2**, the filling material **140** can flow through the guiding area **131b** to squeeze the air between the chip **110** and the flexible substrate **130** out of the chip mounting area **131a**, fill in the first guiding channel **R1** and the second guiding channel **R2** and cover the connective line **134** to enhance the reliability of the semiconductor package **100**.

(13) Because of the flow guiding strips **133** and the first and second guiding channels **R1** and **R2** located between the two adjacent flow guiding strips **133**, the filling material **140** can squeeze out the air between the chip **110** and the flexible substrate **130** and smoothly fill the space between the chip **110** and the flexible substrate **130** to increase filling rate. Moreover, the flow guiding strips **133** are provided to separate the first surface **111** of the chip **110** and the second surface **131** of the flexible substrate **130** such that the contacting between the first surface **111** of the chip **110** and the second surface **131** of the flexible substrate **130** and the flow blocking of the fill material **140** between the chip **110** and the flexible substrate **130** are preventable.

(14) While this invention has been particularly illustrated and described in detail with respect to the preferred embodiments thereof, it will be clearly understood by those skilled in the art that is not limited to the specific features shown and described and various modified and changed in form and details may be made without departing from the scope of the claims.

Claims

1. A semiconductor package comprising: a chip including a first surface and a plurality of conductive pads which are visible from the first surface, a first area and a second area are defined on the first surface, the first area is located between an edge of the first surface and the second area, the plurality of conductive pads are located on the first area; a plurality of bumps each disposed on one of the plurality of conductive pads; and a flexible substrate including a second surface, a plurality of conductive lines and a plurality of flow guiding strips, a chip mounting area, a guiding area and a wire area are defined on the second surface, the guiding area is located inside the chip

mounting area, each of the plurality of conductive lines includes an inner lead and a wire which are connected with each other, the inner lead, the wire and the plurality of flow guiding strips are disposed on the chip mounting area, the wire area and the guiding area respectively along a first direction, the plurality of flow guiding strips are arranged to extend toward the inner lead of the plurality of conductive lines along the first direction and are not connected to the plurality of conductive lines, a first guiding channel is located between the adjacent flow guiding strips along the first direction and a second guiding channel is located between the adjacent flow guiding strips along a second direction which intersects the first direction, the second guiding channel is communicated with the first guiding channel, the chip is mounted on the chip mounting area and bonded to the inner lead of the plurality of conductive lines by the plurality of bumps, the first surface of the chip faces toward the second surface of the flexible substrate, the second area of the first surface of the chip is located above the guiding area of the second surface of the flexible substrate, the plurality of flow guiding strips are located between the second area and the guiding area and configured to separate the first surface of the chip and the second surface of the flexible substrate.

2. The semiconductor package in accordance with claim 1, wherein the flexible substrate includes at least one connective line which is located on the chip mounting area, both ends of the at least one connective line are configured to bridge two of the plurality of conductive lines, the at least one connective line and the inner lead of the plurality of conductive lines surround the guiding area.

3. The semiconductor package in accordance with claim 1 further comprising a filling material, wherein the plurality of flow guiding strips are configured to contact the first surface of the chip, the filling material is configured to be filled between the first surface of the chip and the second surface of the flexible substrate and filled in the first guiding channel and the second guiding channel.

4. The semiconductor package in accordance with claim 2 further comprising a filling material, wherein the plurality of flow guiding strips are configured to contact the first surface of the chip, the filling material is configured to be filled between the first surface of the chip and the second surface of the flexible substrate and filled in the first guiding channel and the second guiding channel.

5. The semiconductor package in accordance with claim 1 further comprising a filling material, wherein there is a gap between the first surface of the chip and each of the plurality of flow guiding strips, the filling material is configured to be filled between the first surface of the chip and the second surface of the flexible substrate and filled in the first guiding channel, the second guiding channel and the gap.

6. The semiconductor package in accordance with claim 2 further comprising a filling material, wherein there is a gap between the first surface of the chip and each of the plurality of flow guiding strips, the filling material is configured to be filled between the first surface of the chip and the second surface of the flexible substrate and filled in the first guiding channel, the second guiding channel and the gap.

7. The semiconductor package in accordance with claim 1, wherein the guiding area has a length along the first direction, the length is greater than or equal to 300 μm .

8. The semiconductor package in accordance with claim 7, wherein the first guiding channel has a first width along the first direction, the first width is greater than or equal to 50 μm and less than or equal to 100 μm .

9. The semiconductor package in accordance with claim 2, wherein the guiding area has a length along the first direction, the length is greater than or equal to 300 μm .

10. The semiconductor package in accordance with claim 9, wherein the first guiding channel has a first width along the first direction, the first width is greater than or equal to 50 μm and less than or equal to 100 μm .

11. The semiconductor package in accordance with claim 1, wherein the guiding area has a width

along the second direction, the width is greater than or equal to 300 μm .

12. The semiconductor package in accordance with claim 11, wherein the width of the guiding area along the second direction is greater than 500 μm , the second guiding channel has a second width along the second direction, the second width is greater than or equal to 100 μm and less than or equal to 500 μm .

13. The semiconductor package in accordance with claim 2, wherein the guiding area has a width along the second direction, the width is greater than or equal to 300 μm .

14. The semiconductor package in accordance with claim 13, wherein the width of the guiding area along the second direction is greater than 500 μm , the second guiding channel has a second width along the second direction, the second width is greater than or equal to 100 μm and less than or equal to 500 μm .

15. The semiconductor package in accordance with claim 1, wherein each of the plurality of flow guiding strips has a length along the first direction, the length is greater than or equal to 100 μm and less than or equal to 300 μm , each of the plurality of flow guiding strips has a width along the second direction, the width is greater than or equal to 7 μm and less than or equal to 30 μm .

16. The semiconductor package in accordance with claim 1, wherein a distance between each of the plurality of flow guiding strips and the inner lead of each of the plurality of conductive lines along the first direction is greater than or equal to 50 μm and less than or equal to 100 μm .

17. The semiconductor package in accordance with claim 1, wherein the plurality of the flow guiding strips and the plurality of conductive lines are made of the same material.

18. The semiconductor package in accordance with claim 1, wherein the flexible substrate further includes a solder resist layer, the solder resist layer is configured to cover the second surface of the flexible substrate and the wire and not cover the inner lead, the guiding area and the chip mounting area, the plurality of flow guiding strips and the solder resist layer are made of the same material.
